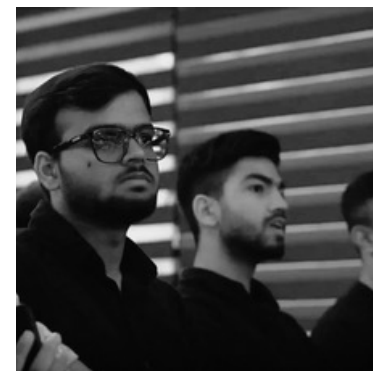
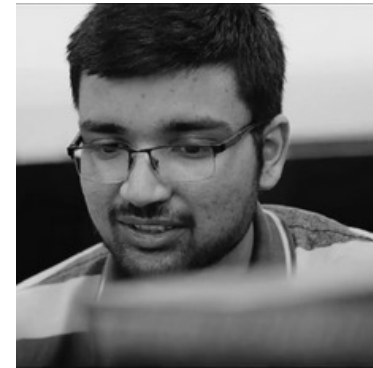


# TechForChange

## COHORT 4





Funded by  
**ciena**

**nasscom**  
foundation



**Adarsh Goyal**  
**Founder**



**Sneha Gupta**  
**Co-Founder**

# PROBLEM STATEMENT

- ▶ Visually impaired students face major barriers in STEM education due to its visual nature, lack of accessible materials, and costly assistive devices.
- ▶ Affordable, tactile-based, software-integrated tools are urgently needed to make STEM concepts accessible.
- ▶ Millions of visually impaired students globally—especially in developing countries like India—are excluded from STEM learning and careers.
- ▶ Leads to educational exclusion, unemployment, and low literacy among visually impaired individuals.



# Market Analysis & Opportunity

## Relevance

Over 43 million people are blind globally, and access to Braille or tactile learning tools remains limited. This lack directly affects education and employment opportunities, especially in developing regions.

## Highlights

The refreshable Braille display market is growing fast — valued around USD 500M in 2024, projected to reach USD 1B by 2033 (8% CAGR). Studies also show Braille literacy strongly improves employment rates among visually impaired individuals.

## Competitors

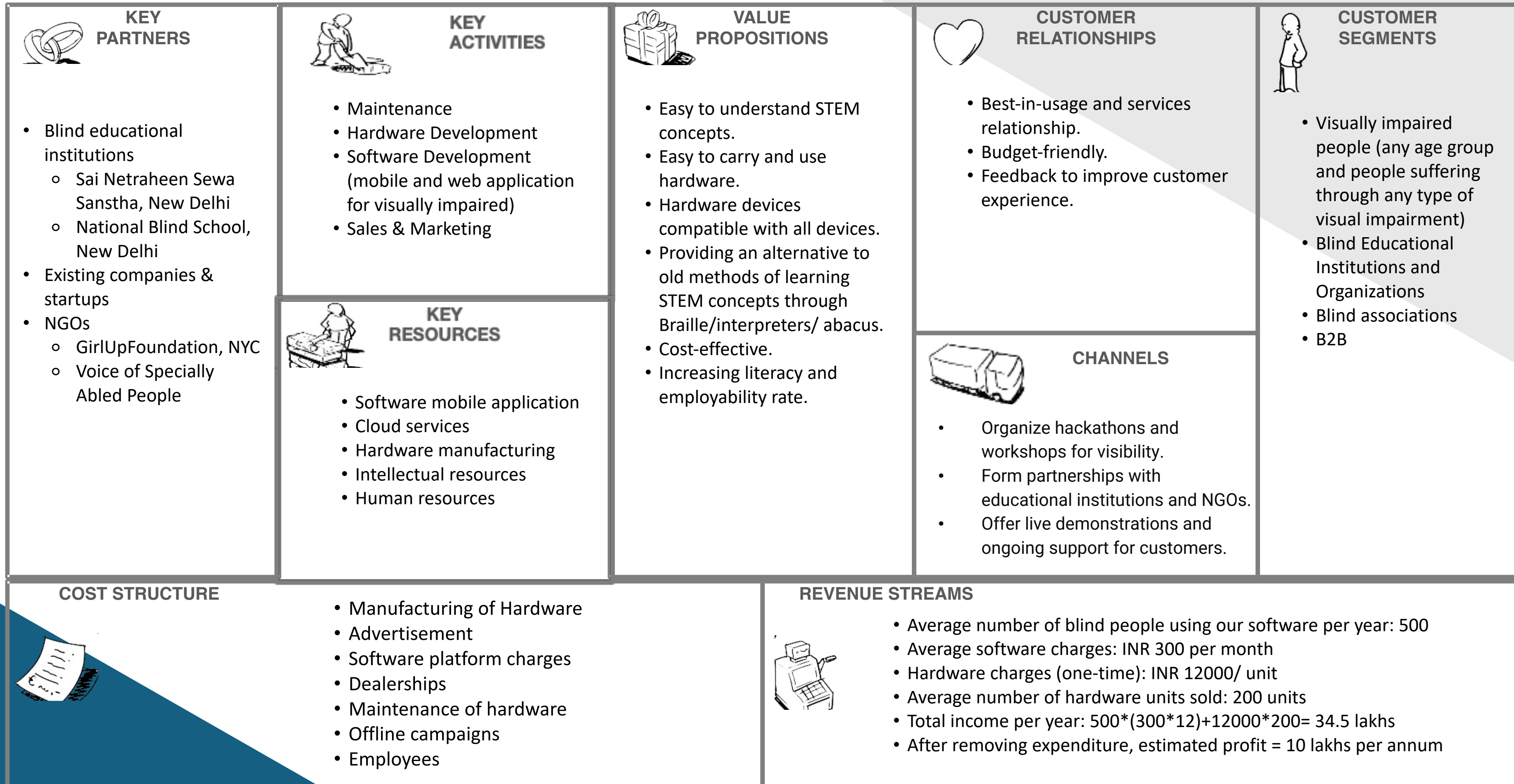
Companies like HumanWare, Orbit Research, Dot Inc., Freedom Scientific, and Innovision (BrailleMe) are active in this space, focusing on making Braille more digital and portable.

## Gaps

- High cost of multi-line/tactile devices
- Limited graphics support for STEM learning
- Lack of local language & training ecosystem
- Low accessibility in developing countries



# Business Model Canvas

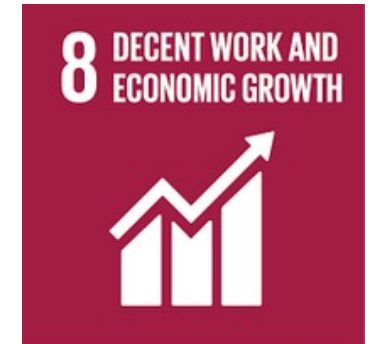


# Proposed solution and SDGs:

A lightweight, modular refreshable Braille display using 5V solenoids, ESP32 microcontroller, I2C architecture, and 3D-printed housing, reducing cost from \$10,000–\$20,000 to \$500–\$1,500.

It directly solves the affordability, multi-line display, and portability gaps—making Braille and STEM content accessible to millions of visually impaired learners and professionals at a fraction of the cost.

Combines low-cost components, open-source scalability, and local 3D printing, enabling rapid manufacturing, local repair, and 19x wider accessibility compared to current products.





# Customer Validation





# Thank You

